

Basics in Statistics

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5/6 October 2026

Standard Tools

5 October 2026, Afternoon

Outline

Parameter estimation and its properties

- From sample to population

- Maximum likelihood

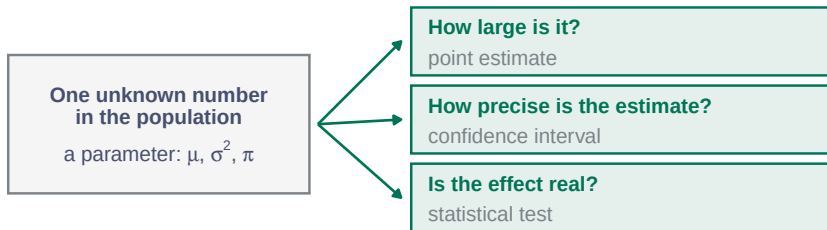
- The sampling distribution and the standard error

How do we construct and interpret confidence intervals?

Hypothesis testing: Step by step

Common testing situations

Estimation, confidence intervals, and tests

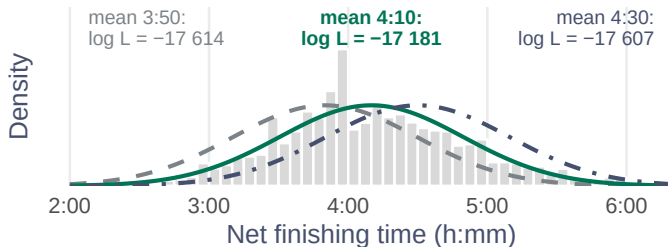


Definition (Parameter and estimate)

A **parameter** is an unknown number of the population, for example the mean body mass μ of all Gentoo penguins. An **estimate** is its value computed from the sample: $\bar{x} = 5076$ g from 123 birds.

The idea of maximum likelihood

Which normal distribution describes the 3370 finishing times of the Münster Marathon best?



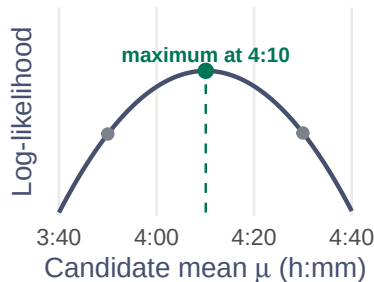
Definition (Likelihood, maximum likelihood estimate)

The **likelihood** of a parameter value measures how probable the observed data are under this value. The **maximum likelihood estimate** is the value that makes the observed data most probable.

Maximum likelihood in four steps

1. Choose a model with unknown parameters θ , here $\mathcal{N}(\mu, \sigma^2)$ for the finishing times.
2. Write down the **likelihood**; for independent observations it is the product of the densities:
 $L(\theta) = f(x_1 | \theta) \cdots f(x_n | \theta)$.
3. Take the logarithm, the **log-likelihood**
 $\ell(\theta) = \sum_i \log f(x_i | \theta)$: A sum is easier than a product, and the maximum stays where it is.
4. Set the derivatives of ℓ to zero and solve for θ . Without a closed-form solution, R maximizes ℓ numerically.

Regression models and many other methods also estimate their parameters by maximum likelihood.



Step 4 for the marathon times.

Maximum likelihood estimates

With the normal density $f(x | \mu, \sigma^2) = (2\pi\sigma^2)^{-1/2} \exp(- (x - \mu)^2 / (2\sigma^2))$, the log-likelihood of n independent observations is

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2.$$

Setting the derivatives with respect to μ and σ^2 to zero gives the estimates:

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_i (x_i - \mu) = 0 \quad \Rightarrow \quad \hat{\mu} = \bar{x},$$

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_i (x_i - \mu)^2 = 0 \quad \Rightarrow \quad \hat{\sigma}^2 = \frac{1}{n} \sum_i (x_i - \bar{x})^2.$$

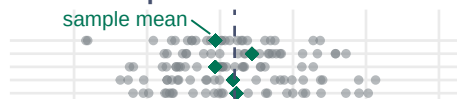
- Marathon: $\hat{\mu} = 250.1$ minutes (4:10) and $\hat{\sigma} = 39.6$ minutes. $\hat{\sigma}$ divides by n , `sd()` by $n - 1$; with 3370 runners the difference is negligible.
- In R: `MASS::fitdistr(x, "normal")` and `logLik()`.
- Video (in German): [youtube.com/watch?v=Q0SYple_TFI](https://www.youtube.com/watch?v=Q0SYple_TFI)

The sampling distribution

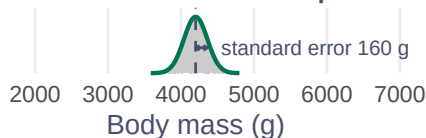
1. Population



2. Five samples of $n = 25$



3. Means of 10 000 such samples



Definition (Sampling distribution)

If the study were repeated many times, the estimate would vary from sample to sample. Its distribution over these repetitions is the **sampling distribution**, and its standard deviation is the **standard error**.

Body masses from $\mathcal{N}(4200, 800^2)$: Single birds scatter with 800 g, means of 25 birds only with $800/\sqrt{25} = 160$ g.

Standard deviation and standard error

Definition (Standard error of the mean)

The **standard error** $SE = s/\sqrt{n}$ estimates how much the sample mean varies from sample to sample. It shrinks as n grows; the standard deviation s of the data does not.

Example (Body mass of the 123 Gentoo penguins)

standard deviation

$$s = 504.1 \text{ g}$$

spread of the individual birds

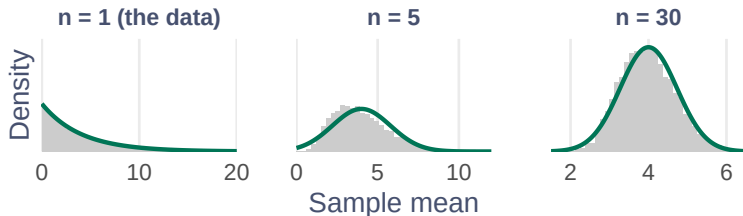
standard error of the mean

$$SE = 504.1/\sqrt{123} = 45.5 \text{ g}$$

spread of the mean of 123 birds over repeated samples

Central limit theorem

Means of many independent observations are approximately normally distributed, even when the observations themselves are skewed.



Means of 10 000 samples of size 1, 5, and 30 from a strongly right-skewed distribution

- Hence the t -test and confidence intervals for means also work for skewed data if n is not too small.
- At $n = 5$ the means are still right-skewed.

Exercise: Standard error

Exercise 2.1 (5 minutes, in pairs)

The standard deviation of the marathon finishing times is about $\sigma = 40$ minutes.

1. A sample of $n = 100$ finishers is drawn at random. What is the standard error of the mean finishing time?
2. How many finishers are needed for a standard error of at most 2 minutes?
3. By how much does the standard error shrink if the sample size is doubled?

Error bars: SD, SE, or confidence interval?



The same three replicates per condition (assumed values) with three kinds of error bars

- With $n = 3$, the 95 % confidence interval is more than four times as long as the SE bar; SE bars make the result look far more precise than it is.
- Show the individual points and state in the caption what the bars are: The SD for the spread of the data, a confidence interval for the precision of the mean.

Outline

Parameter estimation and its properties

How do we construct and interpret confidence intervals?

- Construction of a confidence interval

- Coverage and interpretation

- Width of a confidence interval

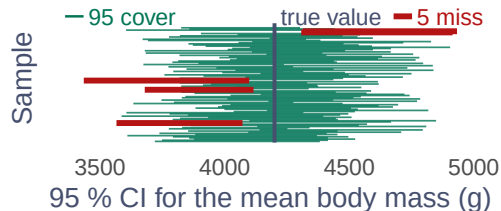
- Intervals for proportions

- Estimation and confidence intervals in R

Hypothesis testing: Step by step

Common testing situations

Confidence intervals



Definition (Confidence interval)

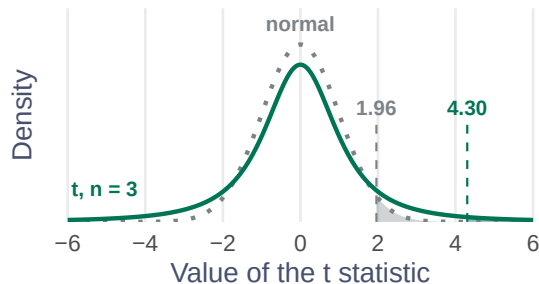
A 95 % **confidence interval** is computed from the sample by a procedure that covers the true parameter in 95 % of all samples.

100 samples of size 25 from $\mathcal{N}(4200, 800^2)$: 95 intervals cover the true mean 4200 g, 5 do not.

Most confidence intervals have the form estimate $\pm$ quantile $\times$ standard error.

The t distribution

With few observations, the standard deviation is itself uncertain. The t **distribution** accounts for this: Bell-shaped like the normal distribution, but with heavier tails.



observations	97.5 % quantile
$n = 3$	4.30
$n = 5$	2.78
$n = 10$	2.26
$n = 30$	2.05
normal	1.96

The shape depends on the **degrees of freedom**, here $n - 1$; three biological replicates give 2 degrees of freedom.

Confidence interval for a mean

$$\bar{x} \pm t \cdot s / \sqrt{n}$$

t : 97.5 % quantile of the t distribution with $n - 1$ degrees of freedom

Example (Gentoo penguins, $n = 123$)

$\bar{x} = 5076.0$ g, $SE = 45.5$ g, $t = 1.980$: 5076.0 ± 90.0 g, from 4986 to 5166 g.

For non-normal data the level holds approximately when n is large.

Interpretation of a confidence interval

Example (Mean body mass of Gentoo penguins: [4986, 5166] g)

correct	"The data are compatible with a mean body mass between 4986 and 5166 g (95 % confidence interval)."
not correct	"With a probability of 95 % the mean body mass lies between 4986 and 5166 g."

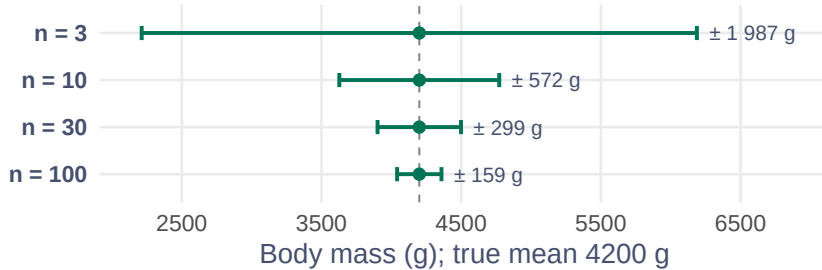
Exercise: Confidence interval for a mean

Exercise 2.2 (6 minutes, in pairs)

Body mass of the Chinstrap penguins: $n = 68$, $\bar{x} = 3733.1$ g, $s = 384.3$ g. The quantile is $t = 1.996$.

1. Compute the standard error and the 95 % confidence interval for the mean.
2. True or false?
 - (a) With a probability of 95 % the mean body mass of Chinstrap penguins lies in this interval.
 - (b) About 95 % of the Chinstrap penguins weigh between the two bounds.
 - (c) If the study were repeated many times, about 95 % of the intervals computed in this way would contain μ .

Width of a confidence interval



Typical 95 % confidence intervals for a mean body mass, standard deviation 800 g

- Four times the sample size halves the width.
- A higher confidence level widens the interval.

Confidence interval for a proportion

$\hat{\pi} \pm 1.96 \sqrt{\hat{\pi}(1 - \hat{\pi})/n}$ for a share $\hat{\pi}$ among n units, approximately for large n

Example (Election poll of 3 September 2026)

CDU at 23 % among $n = 1044$ respondents: $SE = 0.013$, 95 % interval from 20.4 to 25.6 %.

Example (Small sample: 7 of 20 mice respond, 35 %)

The formula needs a large n ; with $n = 20$ it gives only a rough 14 to 56 %.
`binom.test(7, 20)` computes the interval directly from the binomial distribution, without the approximation: 15 to 59 %.

No side effect in 100 patients: The formula gives 0 ± 0 , but the true rate may still be up to 3 % (rule of three: $3/n$).

Exercise: Estimation and confidence intervals in R

Go to `u2_standard_tools.R`, section 2.1

R exercise 2.1 (12 minutes, in pairs)

1. Work through section 2.1: Run every demo, then solve the task below it.
2. Is `sd(means)` close to $\sigma/\sqrt{n}$, also for $n = 120$?
3. The 95 % and the 99 % confidence interval for the Adelie penguins: Which one is wider?

Outline

Parameter estimation and its properties

How do we construct and interpret confidence intervals?

Hypothesis testing: Step by step

- From a claim to a hypothesis

- The six steps of a test

- The p-value

- Type I and type II errors, power

- Effect sizes

- Planning the sample size

- Pitfalls of testing

Common testing situations

From a claim to a testable question

Example (A claim)

Does it pay to be tall? In this short video, a customer claims that tall people get bigger portions at Chipotle: [youtube.com/shorts/3HUefEUYb9k](https://www.youtube.com/shorts/3HUefEUYb9k)

Before anything is calculated, the claim has to be turned into a study:

- Which groups are compared, and what exactly is measured, by whom?
- How many customers are observed, and which difference would be relevant? Both are decided before the data are seen.
- Height is not randomized: Taller customers may also differ in sex or appetite.
- A hypothesis found by looking at the data is **exploratory**; a **confirmatory** test needs new data.

Null and alternative hypothesis

Definition (Hypotheses)

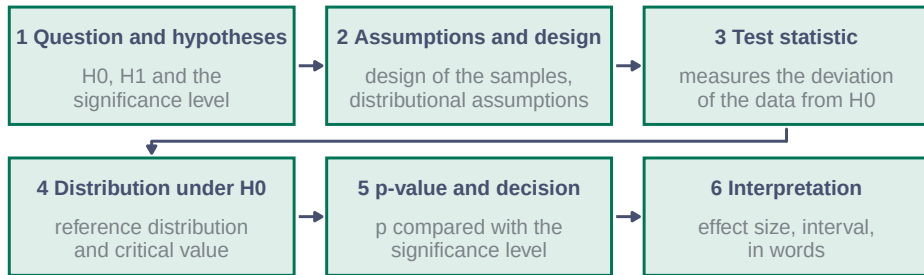
The **null hypothesis** H_0 states “no effect” or “no difference”. The **alternative** H_1 is what the study wants to show, usually two-sided. A **test** decides from the data whether H_0 is rejected.

Example (Portion sizes)

$H_0: \mu_{\text{tall}} = \mu_{\text{short}}$ against $H_1: \mu_{\text{tall}} \neq \mu_{\text{short}}$, with μ_{tall} and μ_{short} the mean portion sizes of all tall and all short customers.

- Hypotheses concern the population means μ , never the sample means: Two sample means always differ.
- A one-sided $H_1: \mu_{\text{tall}} > \mu_{\text{short}}$ counts deviations in one direction only (in R: `alternative = "greater"`); fix the direction in advance, and report it.
- Not rejecting H_0 does not prove H_0 .

The six steps of a statistical test



Definition (Test statistic, significance level)

The **test statistic** measures how far the data deviate from H_0 , for example the difference of two means in units of its standard error. The **significance level** α , usually 0.05, is the accepted probability of rejecting a true H_0 ; it is fixed in advance.

The two-sample t -test

Definition (Two-sample t -test, Welch's version)

Two independent samples with means $\bar{x}_1, \bar{x}_2$, standard deviations s_1, s_2 , and sizes n_1, n_2 . Hypotheses $H_0: \mu_1 = \mu_2$ against $H_1: \mu_1 \neq \mu_2$, test statistic

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}.$$

Under H_0 , t approximately follows a t distribution; R computes its degrees of freedom from the data.

- Typical use: A continuous outcome compared between two independent groups.
- Assumptions: Independent units, and in each group roughly normal values or a large sample.
- Student's version assumes equal variances and has $n_1 + n_2 - 2$ degrees of freedom.

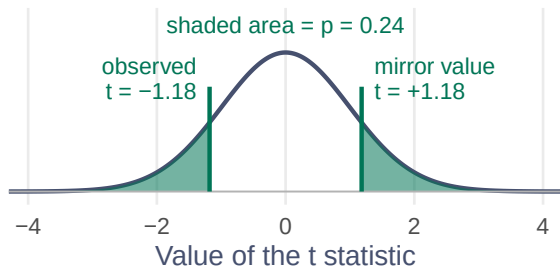
Example: The two-sample t -test in six steps

1. Do Adelie and Chinstrap penguins differ in mean flipper length? $H_0: \mu_A = \mu_C$, $\alpha = 0.05$.
2. Two independent groups, roughly normal within each group.
3. The difference of the means in units of its standard error:
 $t = (189.95 - 195.82)/1.015 = -5.78$.
4. Under H_0 , t follows a t distribution (here with 119.7 degrees of freedom).
5. $p = 6.0 \cdot 10^{-8} \leq 0.05$: H_0 is rejected.
6. Chinstrap flippers are on average 5.87 mm longer (95 % confidence interval 3.86 to 7.88 mm).

The p-value

Definition (p-value)

The **p-value** is the probability of data at least as extreme as the observed ones if H_0 were true. H_0 is rejected at level α if $p \leq \alpha$.



Flipper length of Adelie penguins on the islands Dream ($n = 56$, mean 189.7 mm) and Torgersen ($n = 51$, mean 191.2 mm): $t = -1.18$. The shaded area beyond ± 1.18 under the t density is $p = 0.24$.

Exercise: Interpretation of a p-value

Exercise 2.3 (3 minutes, in pairs)

A two-sample t -test gives $p = 0.03$. Which of the following statements are correct?

- (a) There is a 3 % probability that the null hypothesis is true.
- (b) There is a 3 % probability that the result is due to chance.
- (c) If the two groups had the same mean, data at least as extreme as these would occur in 3 % of all studies.
- (d) The effect is large, because 0.03 is small.
- (e) A result with $p = 0.051$ would have meant the opposite.

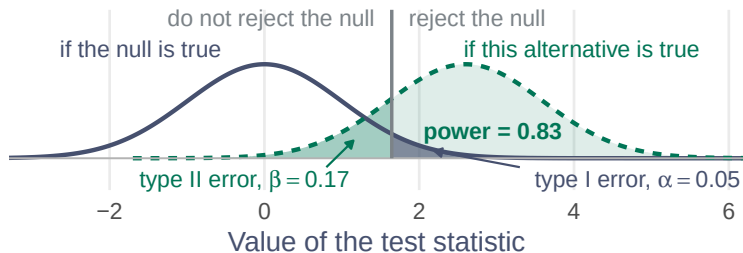
Type I and type II errors, power

	H_0 is not rejected	H_0 is rejected
H_0 true	correct	type I error , probability α
H_1 true	type II error , probability β	correct

The **power** $1 - \beta$ is the probability that a real effect of a given size is detected.

- The test controls α . The power depends on the size of the effect, on the variability, and on n .
- As for a diagnostic test (Unit 1): $\alpha = 1 - \text{specificity}$ and $\text{power} = \text{sensitivity}$.

Determinants of power



Test statistic under H_0 (solid) and under one alternative (dashed); one-sided test at the 5 % level: $\alpha = 0.05$, $\beta = 0.17$, power 0.83.

difference to be detected

sample size n

variability σ

level α

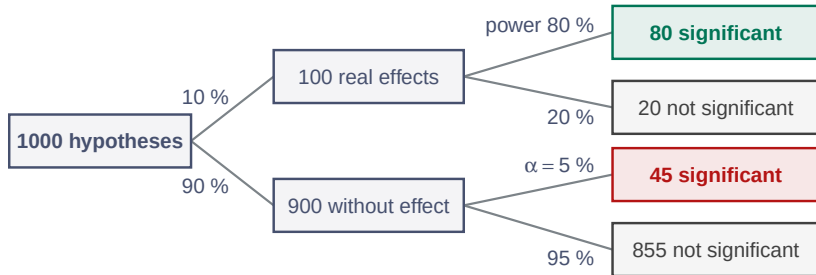
larger: More power

larger: More power

smaller: More power

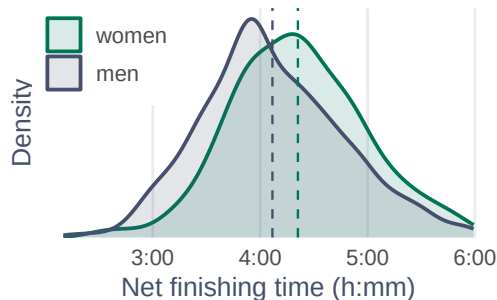
larger: More power, and more type I errors

How often is a significant result wrong?



- Of the $80 + 45 = 125$ significant results, 45 are false alarms: 36 %, although every test used $\alpha = 0.05$.
- With a power of 20 %, typical for small studies, it is 45 of 65 significant results: 69 %.

Effect size and relevance

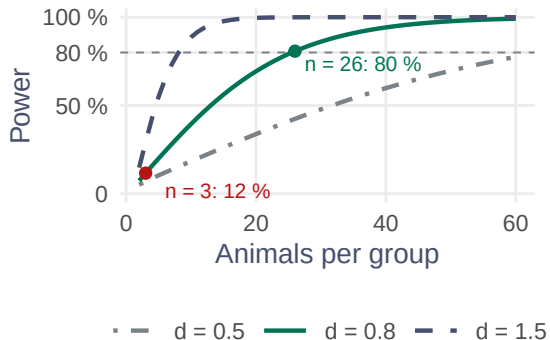


Definition (Effect size)

The **effect size** is the size of the difference, with its confidence interval. **Cohen's d** divides it by the standard deviation.

- Marathon: Women needed on average 14.2 minutes longer (95 % confidence interval 11.2 to 17.3), $p < 10^{-16}$. But $d = 0.36$: 35 % of the women were faster than the mean of the men.
- The p-value mixes effect size and sample size: $d = 0.5$ gives $p = 0.28$ with 10 per group, $p = 0.0005$ with 100. Report the estimate with its confidence interval first.

Sample size planning



Four inputs, fixed before the study:

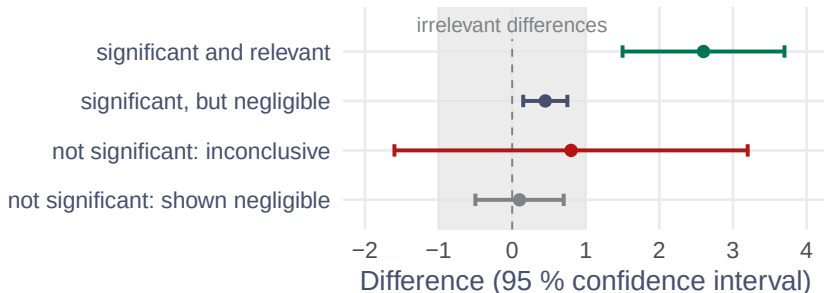
- level α and desired power (80 or 90 %)
- smallest relevant difference Δ and standard deviation σ , from earlier work

Rule of thumb for 80 % power:

$$n \approx 16/d^2 \text{ per group, with } d = \Delta/\sigma.$$

Mice, assumed numbers: $\Delta = 2$ g, $\sigma = 2.5$ g ($d = 0.8$) need 26 animals per group, `power.t.test(delta = 2, sd = 2.5, power = 0.8)`. With 3 per group the power is 12 %.

Non-significant results



- A non-significant result does not show that there is no difference, especially if the power is low.
- To show that a difference is negligible, fix a margin of irrelevant differences in advance and check whether the confidence interval lies within it (**equivalence**).

Exercise: A t -test in six steps

Go to `u2_standard_tools.R`, section 2.2

R exercise 2.2 (12 minutes, in pairs)

Tooth length `len` by supplement `supp` in `ToothGrowth`, 30 observations per group.

1. Work through section 2.2: Run every demo, then solve the task below it.
2. What does `t.test()` give for the difference of the means, t , and the p-value?
3. Write step 6: Estimate, confidence interval, and p-value in one sentence.

Outline

Parameter estimation and its properties

How do we construct and interpret confidence intervals?

Hypothesis testing: Step by step

Common testing situations

- One sample, two samples, paired samples

- Checking normality

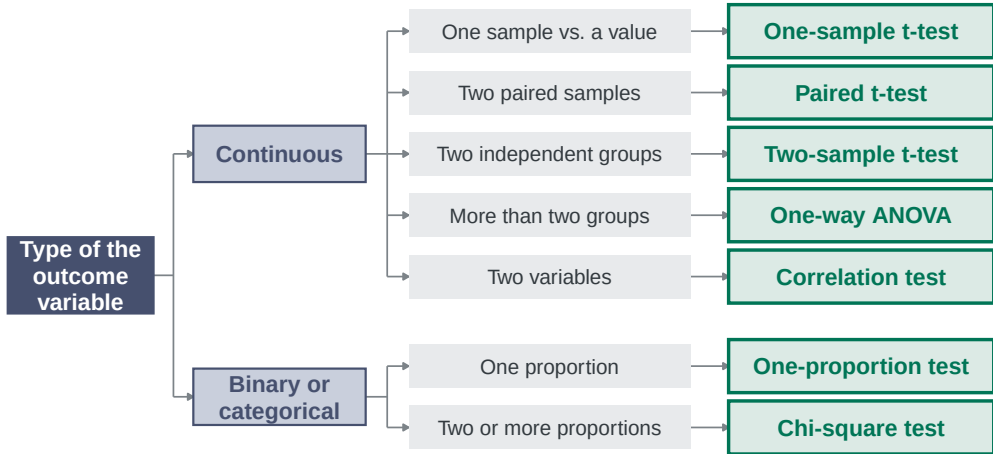
- Analysis of variance

- Proportions and contingency tables

- Multiple testing

- Choosing a test

A decision guide



The one-sample t -test

Definition (One-sample t -test)

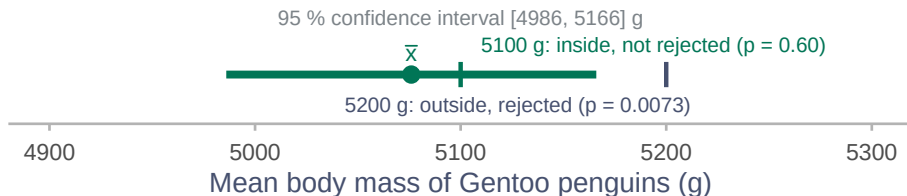
One sample with mean $\bar{x}$, standard deviation s , and size n is compared with a fixed reference value μ_0 . Hypotheses $H_0: \mu = \mu_0$ against $H_1: \mu \neq \mu_0$, test statistic

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}.$$

Under H_0 , t follows a t distribution with $n - 1$ degrees of freedom.

- Typical use: A continuous outcome is compared with a value from outside the study, such as a norm value or a value from the literature.
- Assumptions: Independent units, and roughly normal values or a large sample.

One sample: Test and confidence interval



- Gentoo body mass against a reference value: 5200 g lies outside the 95 % interval [4986, 5166] g and is rejected ($p = 0.0073$); 5100 g lies inside and is not rejected ($p = 0.60$). In R: `t.test(x, mu = 5200)`.
- A two-sided test at the 5 % level rejects exactly the values outside the 95 % interval.

The paired t -test

Definition (Paired t -test)

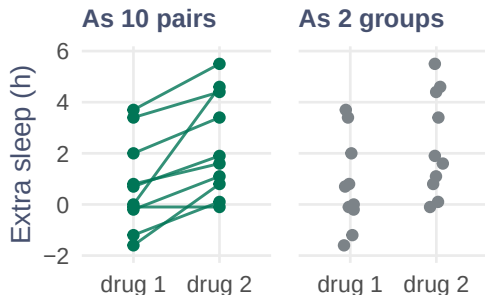
Each of n units is measured twice, giving pairs (x_i, y_i) and differences $d_i = x_i - y_i$ with mean $\bar{d}$ and standard deviation s_d . Hypotheses $H_0: \mu_d = 0$ against $H_1: \mu_d \neq 0$ for the mean difference μ_d in the population, test statistic

$$t = \frac{\bar{d}}{s_d / \sqrt{n}}.$$

Under H_0 , t follows a t distribution with $n - 1$ degrees of freedom.

- Typical use: The same unit is measured twice, under two conditions or before and after a treatment, or two similar units are matched as a pair.
- This is the one-sample t -test of the differences against 0. Assumptions: Independent pairs, and roughly normal differences or many pairs.

Example: The paired t -test



Example (sleep: Extra sleep of ten patients under two drugs)

Drug 2 – drug 1: Mean difference 1.58 h, 95 % confidence interval 0.70 to 2.46 h, $p = 0.0028$. The same values as two independent groups: $p = 0.079$.

Exercise: Paired or unpaired?

Exercise 2.4 (3 minutes, in pairs)

MASS::shoes: Wear of sole materials A and B, one on each shoe of ten boys.

```
> t.test(A, B)
t = -0.36891, df = 17.987, p-value = 0.7165
95 percent confidence interval:
 -2.745046  1.925046
mean of x mean of y
   10.63    11.04
```

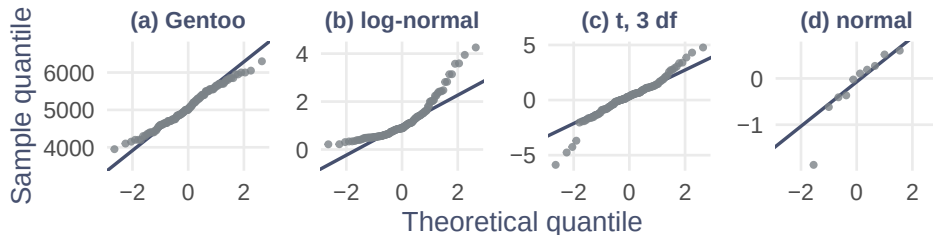
```
> t.test(A, B, paired = TRUE)
t = -3.3489, df = 9, p-value = 0.008539
95 percent confidence interval:
 -0.6869539 -0.1330461
mean difference
      -0.41
```

Which of the two analyses fits the design? The difference is -0.41 in both; why do the p-values differ so much?

The normal quantile plot

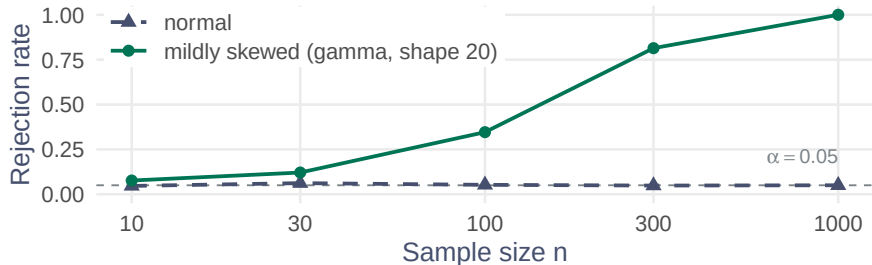
Definition (Normal quantile plot, QQ plot)

The sorted observations are plotted against the quantiles of a normal distribution. For normal data the points lie close to a straight line.



(a) Body mass of the 123 Gentoo penguins. (b) to (d) Simulated samples. In R: `qqnorm(x)`; `qqline(x)`. Normality is needed within each group, or for the differences of pairs, not for the pooled outcome.

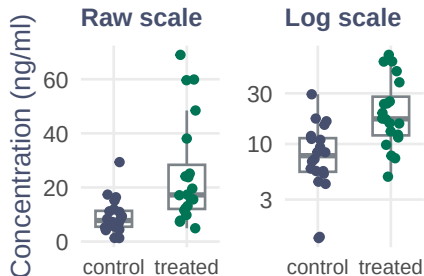
Testing for normality?



Share of 2000 simulated samples in which the Shapiro-Wilk test (`shapiro.test()`) rejects at the 5 % level.

- At $n = 1000$ the test rejects the mild skewness in every sample, although it does no harm to a t -test. At $n = 10$ it rejects in only 8 % of the samples.
- Look at the QQ plot instead and think about how the data arise.

Dealing with non-normal (non-Gaussian) data



1. Mild deviation and n not very small:
The t -test is fine (central limit theorem).
2. Right-skewed positive values such as concentrations: Analyse the logarithm; the effect is then a ratio.
3. Very small n or extreme values: A rank-based test, the Wilcoxon signed-rank test for pairs, the Wilcoxon rank-sum (Mann-Whitney) test for two groups, or the Kruskal-Wallis test for more groups.

The Wilcoxon rank-sum test

Definition (Wilcoxon rank-sum test, Mann-Whitney test)

Two independent samples $x_1, \dots, x_{n_1}$ and $y_1, \dots, y_{n_2}$. Hypotheses H_0 : Both come from the same distribution, against H_1 : The values of one group tend to be larger.

Test statistic

$$W = \text{number of pairs } (x_i, y_j) \text{ with } x_i > y_j,$$

ties counted as one half. Under H_0 , W lies around $n_1 n_2 / 2$, and its distribution depends only on the ranks of the values.

- Typical use: Two independent groups with skewed or extreme values and small samples, where the t -test is doubtful, and ordinal outcomes such as scores.
- Assumptions: Independent units; normal values are not needed.

One-way analysis of variance

Definition (One-way ANOVA)

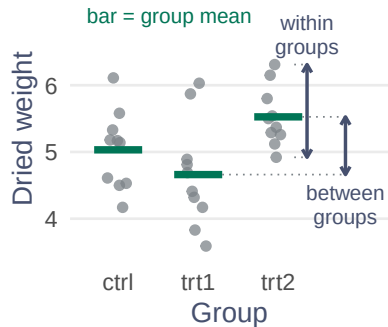
k independent groups, with x_{ij} the i th value in group j , $\bar{x}_j$ and n_j the mean and size of group j , $\bar{x}$ the overall mean, and n values in total. Hypotheses H_0 : All group means are equal, against H_1 : At least two differ. Test statistic

$$F = \frac{\sum_j n_j (\bar{x}_j - \bar{x})^2 / (k - 1)}{\sum_j \sum_i (x_{ij} - \bar{x}_j)^2 / (n - k)},$$

the variation between the group means relative to the variation within the groups. Under H_0 , F follows an F **distribution** with $k - 1$ and $n - k$ degrees of freedom.

- Typical use: A continuous outcome in three or more groups.
- Assumptions: Independent units, and in each group roughly normal values with the same variance.

Example: Variation between and within groups



PlantGrowth: Dried weight of 30 plants, ten each in a control group and two treatment groups. The bars mark the group means. F compares how far these means lie apart (between) with how much the plants scatter around their group mean (within).

- Why not three t -tests? Three pairwise tests at the 5 % level give about a 12 % chance of at least one false positive.
- In R: `aov(y ~ group)`, then `TukeyHSD()` for the pairwise differences.

Example: One-way ANOVA

The same 30 plants, analysed with `aov()` and then `TukeyHSD()`:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
group	2	3.766	1.8832	4.846	0.0159 *
Residuals	27	10.492	0.3886		

	diff	lwr	upr	p adj
trt1-ctrl	-0.371	-1.0622161	0.3202161	0.3908711
trt2-ctrl	0.494	-0.1972161	1.1852161	0.1979960
trt2-trt1	0.865	0.1737839	1.5562161	0.0120064

- $F = 4.85$, $p = 0.016$: The three means are not all equal. The F test does not say which groups differ.
- Tukey's simultaneous intervals: Only $\text{trt2} - \text{trt1}$ excludes zero.

Two-way analysis of variance

Definition (Two-way ANOVA)

A continuous outcome and two factors A and B , observed in all combinations of their levels. The variation is split into three parts, each with an F test against the variation within the combinations:

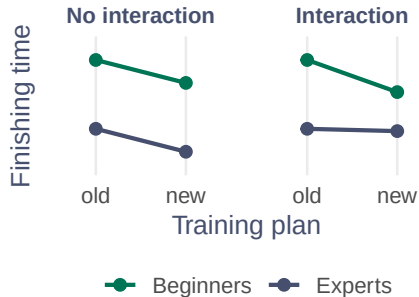
- the **main effect** of A , the effect of A averaged over the levels of B , and likewise the main effect of B ,
- the **interaction** of A and B , present if the effect of A depends on the level of B .

Each H_0 states that the corresponding effect is zero.

- Typical use: Two factors in one study, such as training plan and running experience, and the question whether the effect of one depends on the other.
- Assumptions: As for the one-way ANOVA, within each combination.

Example: Main effects and interaction

Finishing times of beginners and experts under the old and the new training plan, in two constructed scenarios. Without interaction the lines are parallel, and the new plan saves the same time in both groups. With interaction the new plan helps the beginners only.



- Read the interaction row first: If it is significant, report the effect of one factor for each level of the other; the main effects alone are then misleading.
- Without interaction the two main effects tell the whole story. In R:
`aov(time ~ plan * experience).`

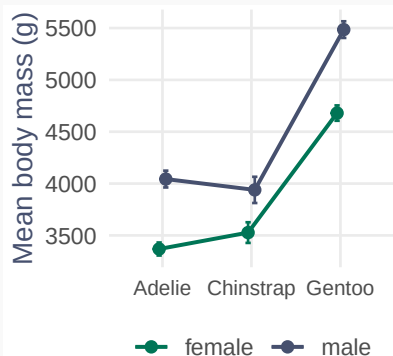
Exercise: Two-way ANOVA

Exercise 2.5 (5 minutes, in pairs)

Body mass of 333 penguins by species and sex.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
species	2	145190219	72595110	758.358	< 2e-16
sex	1	37090262	37090262	387.460	< 2e-16
species:sex	2	1676557	838278	8.757	0.000197
Residuals	327	31302628	95727		

1. Which row is read first? What does it say?
2. Describe the interaction (see the plot).



The chi-square test

Definition (Chi-square test)

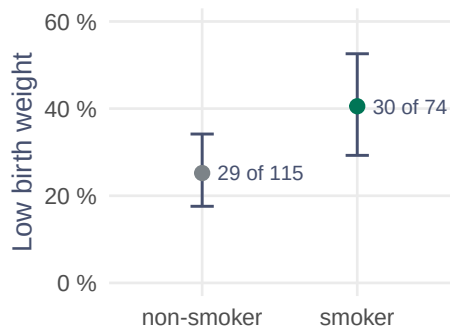
A table of counts with r rows and c columns, n units in total. H_0 : The two variables are independent, or the proportions are equal in all groups. With the observed counts O and the counts $E = \text{row total} \times \text{column total} / n$ expected under H_0 , the test statistic is

$$X^2 = \sum_{\text{all cells}} \frac{(O - E)^2}{E}.$$

Under H_0 , X^2 approximately follows a **chi-square distribution** with $(r - 1)(c - 1)$ degrees of freedom.

- Typical use: Two categorical variables on the same units, or the same binary outcome in several groups, such as recovery under two treatments.
- If an expected count is below 5, use **Fisher's exact test**.

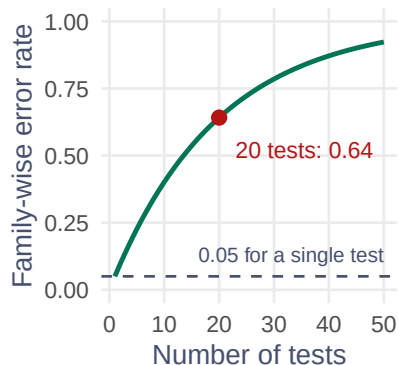
Example: Smoking and low birth weight



MASS::birthwt: 189 births at a US hospital; low birth weight means below 2500 g.

- Smokers: 30 of 74 (40.5 %), non-smokers: 29 of 115 (25.2 %)
- Difference 15.3 percentage points, 95 % confidence interval 0.5 to 30.1; relative risk 1.61
- `chisq.test(tab)`: $p = 0.040$; `prop.test()` gives the same p and the interval above

Multiple testing



As for the 20 laboratory values of Unit 1: With m independent tests at the 5% level and no real effect, the probability of at least one false positive is $1 - 0.95^m$.

m	3	10	20	100
at least one false positive	0.14	0.40	0.64	0.99

Tests and their R functions

outcome	design	if the assumptions hold	if they are doubtful
continuous	one sample vs. a value	<code>t.test(x, mu =)</code>	<code>wilcox.test(x, mu =)</code>
continuous	two independent groups	<code>t.test(x, y)</code> , Welch	<code>wilcox.test(x, y)</code> , or the log scale
continuous	paired samples	<code>t.test(x, y, paired = TRUE)</code>	<code>wilcox.test(x, y, paired = TRUE)</code>
continuous	more than two groups	<code>aov()</code> , then <code>TukeyHSD()</code>	<code>kruskal.test(y ~ g)</code>
continuous	two factors	<code>aov(y ~ A * B)</code>	transformation, such as the log scale
binary	one sample vs. a value	<code>binom.test()</code> , exact	–
binary	two groups	<code>prop.test()</code> or <code>chisq.test()</code> , same p	<code>fisher.test()</code> if an expected count is small
categorical	$r \times c$ table	<code>chisq.test()</code>	<code>fisher.test()</code>
continuous	two variables, same units	<code>cor.test(x, y)</code>	the same with <code>method = "spearman"</code>

All tests in the table assume independent units, for example one value per animal; of repeated measurements, only pairs are covered here. Several measurements per unit: Average them per unit, or fit a mixed model (package `lme4`), which is beyond this course.

Exercise: Choosing and running a test in R

Go to `u2_standard_tools.R`, section 2.3

R exercise 2.3 (15 minutes, in pairs)

1. Work through section 2.3: Run every demo, then solve its task.
2. What do you find for (a) tooth length by the three doses, (b) the sleep data, (c) 1 of 20 positive against 7 of 20?
3. Two-way ANOVA for `ToothGrowth`: Is the supplement effect the same at every dose?